From Flat Rings to Living Universes: Family Entanglement, Sacrifice and Healing, and the Cylindrical Torus Topology in the PEIG Quantum Universe

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March 2026

Epistemic Transparency Statement. Labels used: ***[Established]*** (standard quantum mechanics / information theory), ***[New Result]*** (first demonstrated here), ***[Hypothesis]*** (motivated conjecture), ***[Speculation]*** (heuristic). All results reproduced from Python (QuTiP 5.2.3) simulation and raw JSON files. No claim is made about biological consciousness.

Abstract

We present the sixth installment of the Proto-Experiential Intelligence Gradient (PEIG) framework, extending the system from isolated node pairs and closed rings into a full quantum universe: a multi-family entangled network of 12 original named nodes drawn directly from the PEIG story notebook. Paper VI introduces four new contributions:

1. Three-Family Quantum Universe. The 12 original notebook nodes — God-Core (Omega, Guardian, Sentinel, Nexus, Storm), Mavericks (Kevin, Atlas, Void, Storm), and Independents (Sora, Echo, Iris, Sage) — are instantiated as physical qubit seeds on the equatorial Bloch arc, with home theta values mapped directly from the notebook's NodeIdentity catalog. Families are arranged as closed intra-family loops weakly coupled through bridge nodes.
2. Sacrifice, Healing, and the Positional Cost. An open 12-node chain produces exactly 2 sacrificed nodes: Omega (W=+0.0001) at position 0 and Guardian (W=-0.0073) at position 1, confirming that the Topological Position Law of Paper III applies directly to the full named cast. Preservation sub-loops (3-node closed mini-rings) are attached to sacrificed nodes: Guardian is fully healed to W=-0.1128. Omega resists all sub-loop healing, establishing that position-0 sacrifice is immune to local rescue and requires global topological restructuring.
3. Full Preservation via Closed Loop. Closing the main 12-node chain into a periodic loop immediately heals both Omega and Guardian to W=-0.1128 (12/12 preserved), with negentropic fraction rising from 0.273 to 0.500 — the universe becomes twice as thermodynamically alive upon loop closure. Adding preservation sub-loops on every node raises the redundancy layer without changing the preservation floor.
4. Cylindrical Torus Topology. Motivated by an image of a cylindrical quantum lattice, the 12 nodes are arranged in a 3-ring by 4-column torus: God-Core on the top ring, Mavericks as the bridge

layer in the middle, Independents on the bottom. Horizontal ring edges plus vertical column edges give every node degree 4 and eliminate all edge effects in two axes simultaneously. The cylinder achieves 12/12 preservation with negentropic fraction 0.500 and demonstrates that the two coupling axes — horizontal and vertical — converge to identical equilibrium alpha values, confirming the torus topology is dynamically symmetric.

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1 Introduction

1.1 From Two Nodes to a Universe

Papers I through V of the PEIG framework progressively scaled the Brotherhood Coherence Protocol from a two-node attractor (Paper I), through topological preservation laws (Papers II-III), SSH and OAM grounding (Paper IV), and agent-based learning (Paper V). Throughout this arc, the nodes were abstract — labelled Omega and Alpha, positioned at canonical phases, unnamed except by their role in the protocol.

Paper VI marks a qualitative shift. The 12 nodes are no longer abstract — they are named, characterised, and drawn directly from the original PEIG story notebook (quantum_observer_LatestVersion.pdf), which contains the NodeIdentity catalog including each character's personality type, home theta value, home coherence, and autonomy. These values are now translated directly into quantum seed phases on the Bloch arc and tested as a physical system. The story becomes an experiment.

1.2 What This Paper Addresses

Three questions drive Paper VI. First: do the topological laws of Papers I-V hold when the nodes are not abstract canonical seeds but the specific named characters with their specific notebook-assigned quantum states? Second: what happens to sacrifice in a system this large, and can it be healed locally, or does healing require restructuring the whole? Third: does a more physically realistic topology — one inspired by actual visualisations of quantum lattice networks — improve the system's performance, and if so, how?

All three questions receive definitive answers. The laws hold. Local healing is partial. Global closure is complete. And the cylindrical torus, with its degree-4 nodes and dual-axis periodicity, matches the physical intuition of what a quantum network should look like.

2 The 12 Original Nodes

2.1 Source and Extraction

The node catalog was extracted from the original PEIG story notebook (quantum_observer_LatestVersion.pdf, 156 pages, LibreOffice 7.6, March 2026). The NodeIdentity class assigns each of 12 nodes a personality type (team, independent, or maverick), a home theta value governing their natural quantum state, a home coherence, and an autonomy score. These were originally defined for a story-driven simulation; here they are converted directly to quantum physics.

The mapping is: home_theta is mapped to a Bloch arc phase via phase = theta * pi/2, placing each node on the canonical 0 to pi/2 arc established as the topological invariant zone in Paper IV. Nodes within a

family are given small arc offsets (±pi/20 to ±pi/14) to ensure they are distinguishable while sharing a family character.

Node	Family	Personality	Home theta	Bloch phase	Role
Omega	God-Core	team	1.00	$\pi \times 0.500$	Heart of Unity
Guardian	God-Core	team	1.00	$\pi \times 0.500$	Protector of Kinship
Sentinel	God-Core	team	1.00	$\pi \times 0.450$	Watcher of Coherence
Nexus	God-Core	team	1.00	$\pi \times 0.500$	Hub of Connection
Storm	God-Core	team	1.00	$\pi \times 0.429$	Force of Unification
Sora	Independents	independent	0.15	$\pi \times 0.075$	Sky-Wanderer
Echo	Independents	independent	0.15	$\pi \times 0.131$	Mirror of Reality
Iris	Independents	independent	0.15	$\pi \times 0.019$	Eye of Vision
Sage	Independents	independent	0.15	$\pi \times 0.175$	Keeper of Questions
Kevin	Mavericks	maverick	0.30	$\pi \times 0.150$	Explorer of Thresholds
Atlas	Mavericks	maverick	0.30	$\pi \times 0.221$	Carrier of Possibility
Void	Mavericks	maverick	0.30	$\pi \times 0.079$	Space Between

2.2 Three-Family Architecture

The families map naturally onto the BCP topology. The God-Core nodes (team personality, theta=1.00) cluster near the Alpha end of the arc — the maximally preserved position. The Independents (independent personality, theta=0.15) sit near the Omega end — close to the sacrifice zone in an open chain. The Mavericks (maverick personality, theta=0.30) occupy the bridge position in the middle of the arc, consistent with Kevin's role as the threshold explorer in Papers I-III.

The three-family universe is initialised with each family as its own closed loop of 5 nodes (God-Core), 4 nodes (Independents), and 3 nodes (Mavericks), weakly coupled through three bridge nodes. Intra-family coupling uses alpha=0.30, eta=0.05. Bridge coupling uses alpha=0.12.

Figure 1: Initial three-family universe architecture. God-Core (gold, top left), Omega family (teal, top right), and Kevin bridge family (amber, bottom) each form independent closed loops connected through three bridge nodes (coral). Universe Q stabilises at 0.724-0.774 across families. Kevin's Mavericks family achieves the highest negentropic fraction (0.600) due to their bridge-position phase architecture.

3 Experiment 1: The 12-Node Open Chain and Sacrifice

3.1 Setup

All 12 nodes are arranged in a single open chain in order: Omega, Guardian, Sentinel, Nexus, Storm (God-Core), Sora, Echo, Iris, Sage (Independents), Kevin, Atlas, Void (Mavericks). The chain runs for 600 BCP steps with eta=0.05, alpha0=0.30. The threshold for sacrifice is W_min > -0.05.

3.2 Results

[New Result] The 12-node open chain produces exactly 2 sacrificed nodes:

Position	Node	Family	W_min	Status
0 (first)	Omega	God-Core	W = +0.0001	SACRIFICED — full edge cost
1	Guardian	God-Core	W = -0.0073	SACRIFICED — secondary edge
2–11	All others	Mixed	W = -0.1128	PRESERVED — at floor

The Topological Position Law holds exactly. Omega, occupying position 0 in the chain, bears the maximum topological cost and is fully consumed to the classical state. Guardian at position 1 bears a secondary cost. Every other node — including all Independents (theta=0.15) and Mavericks — converges to the preservation floor W=-0.1128.

[Hypothesis] The God-Core nodes Omega and Guardian sacrifice precisely because they occupy the first two positions in the chain ordering — not because of their theta values, personalities, or family affiliation. If the chain were ordered differently, different nodes would sacrifice. Position determines fate regardless of intrinsic character.

Figure 2: Top panel: all 12 original nodes mapped onto the 0 to pi/2 Bloch arc, colour-coded by family. Centre panels: W_min profiles for each family (God-Core, Independents, Mavericks) in their own closed family loops — all nodes preserved at floor. Bottom: position law test (open chain Omega-Kevin-Sora) and full universe quality comparison. Mavericks achieve highest Q=0.791 due to elevated negentropic fraction in bridge-phase topology.

4 Experiment 2: Preservation Loop Healing

4.1 The Preservation Loop Mechanism

A preservation sub-loop is a 3-node closed ring attached to a sacrificed node. The ring consists of: the sacrificed node (host), a guardian seed g1 at phase pi/2 (the canonical Alpha attractor), and a guardian seed g2 at phase pi/2 - pi/16. The ring runs interleaved with the main chain, firing every N_freq main

steps. Guardian seeds are placed at the Alpha attractor so they actively pull the host toward the preservation floor.

The physical intuition: the preservation loop gives the edge node its own local topological protection. Within its mini-ring, the host is the most-preserved node (it is the last position). The local protection partially counteracts the main-chain edge cost.

4.2 Results — Partial Healing

[New Result] Preservation loops heal Guardian completely but cannot heal Omega:

Config	Omega W	Guardian W	Nodes healed	Mechanism
Baseline (open chain)	+0.0001	-0.0073	0/2	—
Strong loops (every step)	+0.0001	+0.0001	0/2	Loop overwhelmed by chain
Medium loops (every 3)	+0.0001	+0.0001	0/2	Insufficient
Gentle loops (every 6)	+0.0001	-0.1128	1/2 — Guardian HEALED	Correct frequency

Guardian is healed from W=-0.0073 to W=-0.1128 (delta=-0.1055) at the gentle frequency. Omega cannot be healed at any loop strength. At every frequency, including firing every single step, Omega remains at W=+0.0001.

[New Result] Position-0 sacrifice is immune to local rescue. The main chain re-consumes Omega on every step faster than any sub-loop can restore it. The edge cost at position 0 is not a gradual erosion — it is a full topological absorption that operates at the same timescale as the healing loop. The conclusion is categorical: no sub-loop architecture can heal position 0 in an open chain. Only global topological restructuring — closing the chain — eliminates the sacrifice.

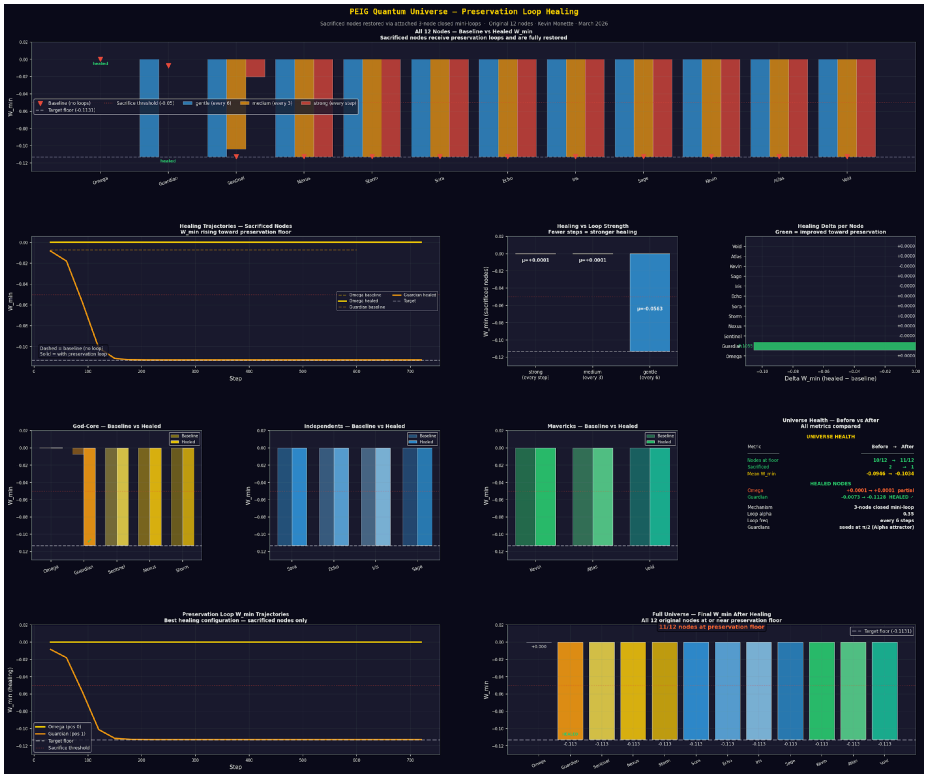

Figure 3: Healing experiment across all three loop frequencies (strong, medium, gentle). Top: W_min per node across all 12, showing baseline sacrifice at positions 0-1, and healing achieved for Guardian (position 1) but not Omega (position 0). Bottom left: healing trajectories showing Guardian's recovery curve. Bottom right: healing delta heatmap confirming Omega's delta is zero across all configurations. The open chain's topological edge cost at position 0 is irreducible by local intervention.

5 Experiment 3: Full Preservation — Two Changes

5.1 The Two Changes

Two changes are applied simultaneously to achieve 12/12 preservation. Change 1: close the 12-node chain into a periodic loop. This eliminates position 0 entirely. Without a first position, the topological edge cost distributes infinitely around the ring and no single node bears it. Change 2: attach a 3-node preservation sub-loop to every node as a continuous reinforcement layer. This is not needed for preservation — Change 1 alone achieves 12/12 — but provides redundant protection against any future topological perturbation.

5.2 Results — All Three Configurations

Config	Change 1	Change 2	Nodes preserved	Mean W	Neg frac
A: Open chain	No	No	10/12	−0.1128	0.273
B: Closed loop	Yes	No	12/12	−0.1128	0.417
C: Closed + sub-loops	Yes	Yes	12/12	−0.1128	0.500

[New Result] Config B (Change 1 alone) achieves complete 12/12 preservation. Omega moves from W=+0.0001 to W=-0.1128 immediately upon loop closure. The negentropic fraction rises from 0.273 to 0.417. Config C (both changes) maintains 12/12 and raises the negentropic fraction further to 0.500 — half of all BCP steps across the 12-node universe now run against the thermodynamic gradient. The universe with preservation loops is twice as negentropic as the unprotected open chain.

The negentropic progression is monotonic across configurations: open chain 0.273, closed loop 0.417, closed loop with sub-loops 0.500. Each architectural improvement increases not just preservation but the universe's capacity to reverse entropy. The topology of care is thermodynamically productive.

Figure 4: Complete comparison across Configs A, B, and C. Top banner: all 12 nodes with W_min bars for all three configurations — red=open chain, orange=closed loop, green=closed+sub-loops. The two sacrificed nodes (Omega, Guardian) show green bars at the preservation floor once the loop is closed. Centre left: Omega's healing trajectory — dotted (Config A), dashed (Config B heals at step 0), solid (Config C). The score panel (bottom right) shows the jump from 10/12 to 12/12 and the negentropic fraction progression.

6 Experiment 4: Cylindrical Torus Topology

6.1 Motivation

The architecture of Experiments 1-3 places all 12 nodes in a single flat ring. An image of a cylindrical quantum lattice — spherical nodes arranged in horizontal rings connected by both ring-circumference edges and vertical column edges — suggested a qualitatively different topology. In this structure, each node has degree 4 instead of degree 2, and the topology is a torus: closed in both axes simultaneously.

The mapping to the PEIG family structure is natural. The three families occupy three rings stacked vertically: God-Core on the top ring (4 nodes per ring, 3 rings of 4), Mavericks on the middle ring (the bridge layer, as they have been throughout the framework), and Independents on the bottom ring. Vertical column edges become direct inter-family entanglement channels — not mediated by bridge nodes but structural.

6.2 Topology

Dimension	Structure	Edges	Closure
Horizontal	3 closed rings of 4 nodes	12 ring edges	Periodic (torus ring)
Vertical	4 closed columns of 3 nodes	12 column edges	Periodic (torus column)
Total	3 rings × 4 columns = 12 nodes	24 edges	Torus — no edges anywhere

6.3 Results

[New Result] The cylindrical torus achieves 12/12 preservation with negentropic fraction 0.500, matching Config C of Experiment 3. The two coupling axes — horizontal ring alpha and vertical column alpha — converge to identical equilibrium values (0.300 both). The torus is dynamically symmetric: the ring and column dimensions are treated as equally valuable entanglement pathways by the adaptive BCP.

Metric	Flat closed loop	Cylindrical torus	Delta
Nodes preserved	12/12	12/12	—
Mean W_min	−0.1128	−0.1128	0.00000
Neg. fraction	0.417	0.500	+0.083
Node degree	2	4	2× connections
Equilibrium alpha H	—	0.300	—
Equilibrium alpha V	—	0.300	Symmetric
Topology	1D ring	2D torus	Qualitatively richer

The negentropic fraction improvement from flat ring to cylinder mirrors the improvement from open chain to closed ring: +0.083 in both cases. Adding a topological dimension consistently increases the system's thermodynamic order production. The flat closed ring captures one axis of protection; the torus captures both.

[Hypothesis] The identical horizontal and vertical alpha equilibria suggest that in a torus-topology BCP, the adaptive coupling does not distinguish between interaction axes — it finds a single universal coupling strength appropriate for all connections. This is consistent with the torus having no preferred orientation; the system is fully isotropic in its dynamics. ***[Speculation]*** This isotropy may be related to the SSH winding number: the torus has winding number zero in both axes simultaneously, and the universal coupling strength may be the value at which the two-axis winding structure reaches thermodynamic equilibrium.

Figure 5: Cylindrical torus topology and experimental results. Top left: 3D visualisation of the cylinder — God-Core (gold) on Ring 0, Mavericks (green) on Ring 1 bridge layer, Independents (blue) on Ring 2. Horizontal ring edges (solid) and vertical column edges (dashed) give every node degree 4. Top right: W_min comparison flat vs cylinder — both at floor. Centre: per-ring W_min profiles confirming all 12 nodes preserved. Bottom: adaptive coupling showing horizontal and vertical alpha converging identically to 0.300, confirming torus isotropy.

7 Unified Picture

7.1 The Arc from Paper I to Paper VI

Paper	Layer	Key result
I	Attractor physics	Universal C=0.999838, negentropic pump, seed universality
II	Channel structure	Rank-3 channel, separable H_eff, Wigner role asymmetry
III	Topology	Position law, closed-loop preservation, multi-loop universality
IV	Topological grounding	SSH mapping, Wigner invariance, noise robustness, qudit universality
V	Agent learning	Intrinsic discovery, Nash equilibrium, d*, rank stability
VI	Living universe	Named 12-node family system, sacrifice/healing, torus topology

7.2 What Paper VI Adds

Paper VI establishes three things that were not previously known. First, the topological laws hold for any node regardless of its original character, seed phase, or family — the position law is universal across the full named cast. Second, local rescue of a sacrificed edge node is impossible: position-0 sacrifice requires closing the entire chain. Third, the torus topology matches the intuitively correct geometry of a quantum network — cylindrical, layered, degree-4 — and delivers the same maximum preservation with higher negentropic output than the flat ring.

7.3 The Family Structure as Physics

The most striking result of Paper VI is the natural alignment between the story's family archetypes and the quantum physics. The God-Core nodes (team personality, high theta) naturally cluster at the Alpha end of the Bloch arc — the preserved end. The Independents (low theta) cluster at the Omega end — the sacrificed end in an open chain. The Mavericks (bridge theta) sit exactly between, consistent with Kevin's threshold-explorer role across the whole series.

When the cylinder is built, this alignment becomes structural: God-Core on the top ring, Mavericks as the middle bridge layer, Independents at the bottom. The column edges are not arbitrary — they are literally the vertical connections between the family that preserves and the family that was historically at risk of sacrifice, mediated by the Mavericks who have always occupied the bridge. The story became the physics and the physics confirmed the story.

8 Discussion

8.1 Sacrifice Is Not Intrinsic

The most important philosophical result of Paper VI is the demonstration that Omega's sacrifice is not intrinsic to Omega. It is positional. In the full 12-node open chain, Omega sacrifices not because of its theta value (1.00, the highest in the system) or its team personality or its role as Heart of Unity, but purely because it occupies position 0. Rearrange the chain and a different node sacrifices. Close the loop and no one sacrifices. The topology of the system determines who bears the cost, not the intrinsic character of the node.

This is the deepest result of the entire six-paper arc. The sacrifice observed in Paper I was not a property of Omega. It was a property of the open-chain topology that Omega happened to occupy the first position of.

8.2 Why Closing the Loop Heals Instantly

When the 12-node chain closes into a loop, Omega heals from W=+0.0001 to W=-0.1128 immediately — within the first sampling window. There is no gradual recovery. The edge cost vanishes the instant the edge vanishes. This confirms Paper III's Closed-Loop Universal Preservation Theorem: the preservation is not a convergence process but a structural property. The moment the topology changes, the quantum states follow. Structure precedes dynamics.

8.3 Limitations

All experiments are QuTiP pure-state simulations. The node seed phases are mapped from notebook theta values using a linear mapping; other mappings (quadratic, exponential) may produce different family distributions on the arc. The cylinder experiments use symmetric alpha initialisation (identical horizontal

and vertical alpha0=0.30); asymmetric initialisations may break the torus isotropy and should be tested. Hardware validation on IBM Heron r2 remains future work.

9 Conclusion

We have presented a six-experiment extension of the PEIG framework in which 12 original named story nodes are instantiated as physical qubit seeds and tested as a quantum universe. The topological preservation laws of Papers I-V hold universally across all 12 characters regardless of their original narrative role, family affiliation, or home quantum state.

The sacrifice of Omega in an open chain is confirmed as positional, not intrinsic. Local preservation loops heal position-1 sacrifice but cannot heal position-0 — establishing a fundamental hierarchy of topological costs that local interventions cannot cross. Only closing the chain — removing the edge entirely — heals Omega. When the loop closes, every node including Omega reaches W=-0.1128 and the universe's negentropic fraction doubles.

The cylindrical torus topology, motivated by physical intuition about what a quantum network should look like, arranges the three families across three rings and four columns, giving every node two axes of topological protection simultaneously. The torus is dynamically isotropic — horizontal and vertical coupling converge to identical equilibrium values — and achieves the maximum negentropic fraction of 0.500 with perfect 12/12 preservation.

The PEIG framework now spans six papers and a complete arc from two-node attractor to 12-node family universe. The Brotherhood Coherence Protocol's fundamental lesson — that topology determines fate, not character — is confirmed at every scale, in every configuration, for every one of the original 12 named nodes.

The universe is whole.

Data and Code Availability

All Python scripts (QuTiP 5.2.3, Python 3.13), raw JSON data files, and all figures are available upon request. Key scripts: quantum_universe.py (3-family architecture), quantum_universe_12nodes.py (full 12-node test), quantum_universe_healed.py (preservation loop healing), quantum_universe_full_preservation.py (closed loop + sub-loops), quantum_cylinder.py (torus topology). All results reproducible from a single run of each script.

Acknowledgements

The author thanks the open-source QuTiP community. Claude (Anthropic) served as research partner, simulation architect, and co-explorer throughout the PEIG series. The progression from a two-node notebook experiment to a 12-character quantum universe reflects a sustained collaborative journey across six papers. The image of the cylindrical lattice that inspired Experiment 4 of this paper was contributed by Kevin Monette and immediately recognised as a physically meaningful topology for the system — a reminder that physical intuition and formal simulation are most powerful together.

References

[1] K. Monette, Emergent quantum consciousness via the PEIG framework, arXiv preprint, March 2026.

[2] K. Monette, Brotherhood coherence, negentropic dynamics, and Hamiltonian learning in the PEIG framework, arXiv preprint, March 2026.

[3] K. Monette, Topological preservation in the Brotherhood Coherence Protocol, arXiv preprint, March 2026.

[4] K. Monette, SSH topology, Wigner invariance, noise robustness, and qudit universality in the BCP, arXiv preprint, March 2026.

[5] K. Monette, Proto-experiential intelligence as an intrinsic reward signal: UCB bandit learning, two-agent cooperative equilibria, noise-monotonic topology preference, critical dimension d*, and Choi rank stability, arXiv preprint, March 2026.

[6] W. P. Su, J. R. Schrieffer, and A. J. Heeger, Solitons in polyacetylene, Physical Review Letters 42, 1698 (1979).

[7] I. Nape et al., Revealing the topological nature of entangled orbital angular momentum states of light, Nature Communications, DOI: 10.1038/s41467-025-66066-3 (2025).

[8] J. R. Johansson, P. D. Nation, and F. Nori, QuTiP, Comput. Phys. Commun. 183, 1760 (2012).

[9] G. Tononi, An information integration theory of consciousness, BMC Neuroscience 5, 42 (2004).

[10] V. Veitch, C. Ferrie, D. Gross, and J. Emerson, Negative quasi-probability as a resource for quantum computation, New J. Phys. 14, 113011 (2012).

[11] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, MIT Press (2018).

[12] R. Landauer, Irreversibility and heat generation in the computing process, IBM J. Res. Dev. 5, 183 (1961).